

Applied Linear Algebra (EE 635): Problem Set 6

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Questions

1. Let V and W be finite-dimensional vector spaces over $\mathbb{R}$. Further, let

$$T : V \rightarrow W$$

be a surjective linear map. Prove that there exists a linear map

$$S : W \rightarrow V$$

such that

$$T \circ S : W \rightarrow W$$

is identity. (In other words, T admits a right inverse.)

2. In Exercises (a)-(d), let $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2\}$ and $\mathcal{C} = \{\mathbf{c}_1, \mathbf{c}_2\}$ be bases for $\mathbb{R}^2$. In each exercise, find the change-of-coordinates matrix from $\mathcal{B}$ to $\mathcal{C}$ and the change-of-coordinates matrix from $\mathcal{C}$ to $\mathcal{B}$.

(a) $\mathbf{b}_1 = \begin{bmatrix} 7 \\ 5 \end{bmatrix}, \mathbf{b}_2 = \begin{bmatrix} -3 \\ -1 \end{bmatrix}, \mathbf{c}_1 = \begin{bmatrix} 1 \\ -5 \end{bmatrix}, \mathbf{c}_2 = \begin{bmatrix} -2 \\ 2 \end{bmatrix}$

(b) $\mathbf{b}_1 = \begin{bmatrix} -1 \\ 8 \end{bmatrix}, \mathbf{b}_2 = \begin{bmatrix} 1 \\ -5 \end{bmatrix}, \mathbf{c}_1 = \begin{bmatrix} 1 \\ 4 \end{bmatrix}, \mathbf{c}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

(c) $\mathbf{b}_1 = \begin{bmatrix} -6 \\ -1 \end{bmatrix}, \mathbf{b}_2 = \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \mathbf{c}_1 = \begin{bmatrix} 2 \\ -1 \end{bmatrix}, \mathbf{c}_2 = \begin{bmatrix} 6 \\ -2 \end{bmatrix}$

(d) $\mathbf{b}_1 = \begin{bmatrix} 7 \\ -2 \end{bmatrix}, \mathbf{b}_2 = \begin{bmatrix} 2 \\ -1 \end{bmatrix}, \mathbf{c}_1 = \begin{bmatrix} 4 \\ 1 \end{bmatrix}, \mathbf{c}_2 = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$

3. In $\mathbb{P}_2$ (the vector space of all polynomials of degree at most 2 with real coefficients), find the change-of-coordinates matrix from the basis

$$\mathcal{B} = \{1 - 2t + t^2, 3 - 5t + 4t^2, 2t + 3t^2\}$$

to the standard basis

$$\mathcal{C} = \{1, t, t^2\}.$$

Then find the $\mathcal{B}$ -coordinate vector for $-1 + 2t$.

4. Let $\mathcal{B}$ and $\mathcal{C}$ be bases for a vector space V . Mark each statement True or False. Justify each answer.
- (a) The columns of the change-of-coordinates matrix $P_{\mathcal{C} \leftarrow \mathcal{B}}$ are $\mathcal{B}$ -coordinate vectors of the vectors in $\mathcal{C}$.
 - (b) If $V = \mathbb{R}^n$ and $\mathcal{C}$ is the standard basis for V , then $P_{\mathcal{C} \leftarrow \mathcal{B}}$ is the same as the change-of-coordinates matrix $P_{\mathcal{B}}$.
 - (c) The columns of $P_{\mathcal{C} \leftarrow \mathcal{B}}$ are linearly independent.
 - (d) If $V = \mathbb{R}^2$, $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2\}$, and $\mathcal{C} = \{\mathbf{c}_1, \mathbf{c}_2\}$, then row reduction of $[\mathbf{c}_1 \ \mathbf{c}_2 \ \mathbf{b}_1 \ \mathbf{b}_2]$ to $[I \ P]$ produces a matrix P that satisfies $[\mathbf{x}]_{\mathcal{B}} = P[\mathbf{x}]_{\mathcal{C}}$ for all $\mathbf{x}$ in V .
5. Exercises a-c concern an $m \times n$ matrix A and what are often called the fundamental subspaces determined by A .
- a. Which of the subspaces $\text{Row } A$, $\text{Col } A$, $\text{Nul } A$, $\text{Row } A^T$, $\text{Col } A^T$, and $\text{Nul } A^T$ are in $\mathbb{R}^m$ and which are in $\mathbb{R}^n$? How many distinct subspaces are in this list?
 - b. Justify the following equalities:
 - i. $\dim(\text{Row } A) + \dim(\text{Nul } A) = n$ (Number of columns of A)
 - ii. $\dim(\text{Col } A) + \dim(\text{Nul } A^T) = m$ (Number of rows of A)
 - c. Use Exercise (b) to explain why the equation $A\mathbf{x} = \mathbf{b}$ has a solution for all $\mathbf{b}$ in $\mathbb{R}^m$ if and only if the equation $A^T\mathbf{x} = \mathbf{0}$ has only the trivial solution.

6. Let

$$T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

be a linear transformation. Consider the basis

$$B = \{v_1 = (1, 1, 0), v_2 = (0, 1, 1), v_3 = (1, 0, 1)\}.$$

Suppose the action of T on the basis vectors of B is defined as

$$T(v_1) = 2v_1 + v_2, \quad T(v_2) = v_2 + v_3, \quad T(v_3) = 3v_1.$$

Determine the matrix representation of T with respect to the basis B .